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Production Uplift in Aging Wells: A Field-Proven Automated Tubing Purging Solution for Brownfield Optimization

U. V. Nair, M. M. Goswami, P. P. Saikia, D. Khound, and S. H. Mehboob, Production Engineering Department, Oil India Limited, Assam, India

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Abstract

In mature oil fields, operational excellence is generally defined by the ability to sustain production, reduce intervention, and extract more value with existing inventory. This case study presents a mature exploratory well, completed in 1994, that faced intermittent flow losses despite a high production rate of 100 KLPD at 50% BS&W. With operational constraints including back- pressure sensitivity and open-hole completion, traditional approaches were ineffective or cost- prohibitive.

Detailed well study revealed increased fluid column density caused by emulsion blockages and water segregation-symptoms of inadequate aeration in the vertical conduit. An initial manual bean-up strategy, though partially effective, demanded high human effort and lacked long-term sustainability. To overcome these challenges and drive performance continuity, a tailored, automated tubing purging system was implemented. Designed for autonomous operation, the system initiates bean-up cycles based on realtime flow conditions and reverts to normal operation once stabilized-eliminating manual intervention and enabling consistent, optimized lift performance.

Key operational outcomes included better vertical lift performance in the well, increase in flowing tubing head pressure from 1718 kg/cm² to 63-64 kg/cm², 10% increase in daily crude oil production, 400% increase in daily natural gas production, zero additional CAPEX, utilizing minimal in-house inventory. This solution exemplifies how automation, data-driven intervention, and lean operational strategies can unlock value from aging wells. It enhances production efficiency, reduces downtime, and minimizes operational risk. It further demonstrates how scalable, field-proven solutions can optimize aging wells while aligning with longterm performance and sustainability goals in upstream operations.

This paper presents a low-complexity, field-deployable automation approach that delivered measurable production gains in a mature, back-pressure-sensitive well- without requiring high-end digital infrastructure or capital investment. By showcasing how simple, scalable modifications can achieve autonomous operation and unlock significant value, the paper offers a replicable model for brownfield optimization. The solution has since been successfully implemented in three additional wells, further validating its adaptability and impact. It fills a critical gap between costly digital overhauls and traditional manual interventions-providing a realistic path to operational excellence for aging wells.

Introduction

As oil and gas reservoirs mature, maintaining stable and economically viable production becomes an increasing challenge for operators worldwide. Declining reservoir pressures, changing fluid properties, and the onset of issues such as emulsion blockages and fluid loading can severely impact the performance of aging wells. These problems are further compounded in wells with openhole completions and high basic sediment and water (BS&W) percentages, where traditional artificial lift interventions or mechanical unloading methods may be either technically unsuitable or economically unjustifiable.

In brownfield environments, maximizing value from existing assets while minimizing capital expenditure is critical to sustaining operations. This is especially true in fields where infrastructure has been in place for decades and where wells are highly sensitive to back-pressure changes, making conventional intervention strategies unreliable or cost-prohibitive.

This paper presents a case study of well M001—a self-flowing exploratory oil well completed as an open completion, as illustrated in Fig-1, in 1994 in the Barail 4th + 5th Sand. **At the time of intervention, the well was producing approximately 104 KLPD with 50% BS&W and a flowing tubing head pressure (FTHP) of 18-19 kg/cm². Annulus-A pressure was recorded at 170 kg/cm².** Over time, the well exhibited signs of production instability, including frequent flow interruptions, heading, and a declining FTHP trend. Conventional manual drawdown methods, though occasionally effective, required frequent human intervention and offered no sustainable solution.

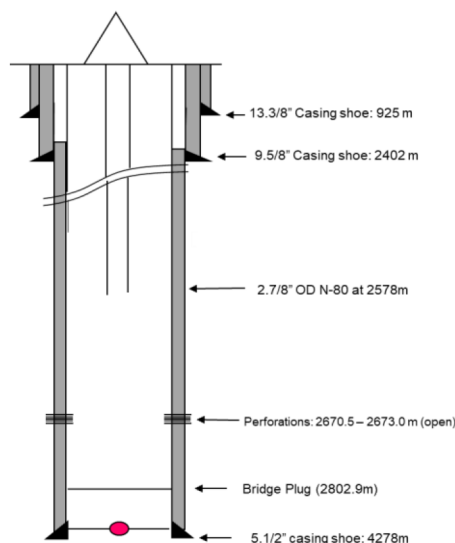


Figure 1—Completion Diagram for M001 (First candidate well for ATPS).

To address these challenges and align with operational excellence principles, a tailored **Automated Tubing Purging Setup (ATPS)** was designed and implemented. The system provided a cost-effective, autonomous means to stabilize production by counteracting emulsion blockages and fluid accumulation, restoring vertical lift performance without additional capital-intensive equipment. **Following its success in M001, the setup was replicated in three additional wells—M002, M025 and M027—where similar production uplift was observed.** This case study demonstrates how a simple, field-driven solution can deliver measurable improvements in production efficiency, reduce operational risk, and offer a replicable model for other mature wells facing similar challenges.

Field Background and Problem Statement

Well M001, producing from one of Oil India Limited's mature oilfields, was completed in January 1994 in the Barail 4th and 5th Sand formations. The well was completed as an open-hole completion and had been self-flowing since inception. **At the time of system implementation, the well was producing approximately 104 KLPD at 50% BS&W, with a flowing tubing head pressure (FTHP) of 18-19 kg/cm², an annulus-A pressure of 170 kg/cm², and an estimated bottom-hole pressure of 228 kg/cm².** As the reservoir continued to deplete, the well began to exhibit symptoms common to late-life production behaviour. Notably, the well experienced:

- A declining trend in FTHP as shown in the graph below in Fig-3
- Frequent flow interruptions and sluggish output
- Evidence of heading due to annulus gas accumulation
- Emulsion blockage formation near the wellbore as shown in the representative image in Fig-2
- A low GOR profile, further limiting natural aeration
- Increasing sensitivity to back-pressure, making stable flow difficult to maintain

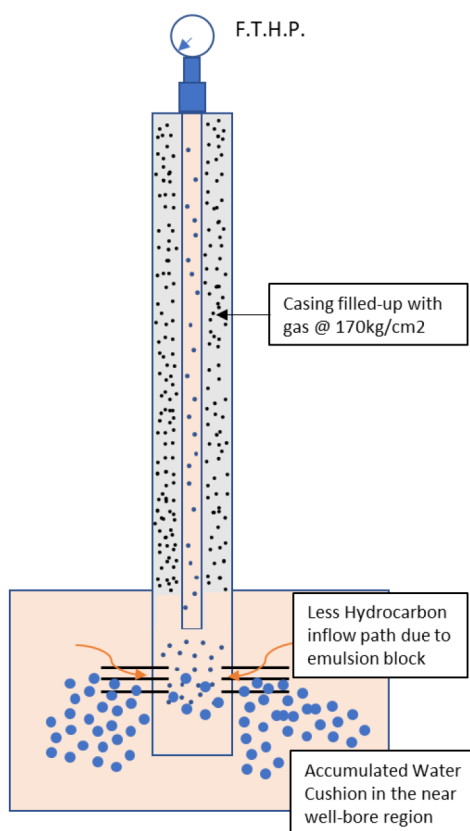


Figure 2—Representative diagram to illustrate well M001's condition prior to ATPS installation.

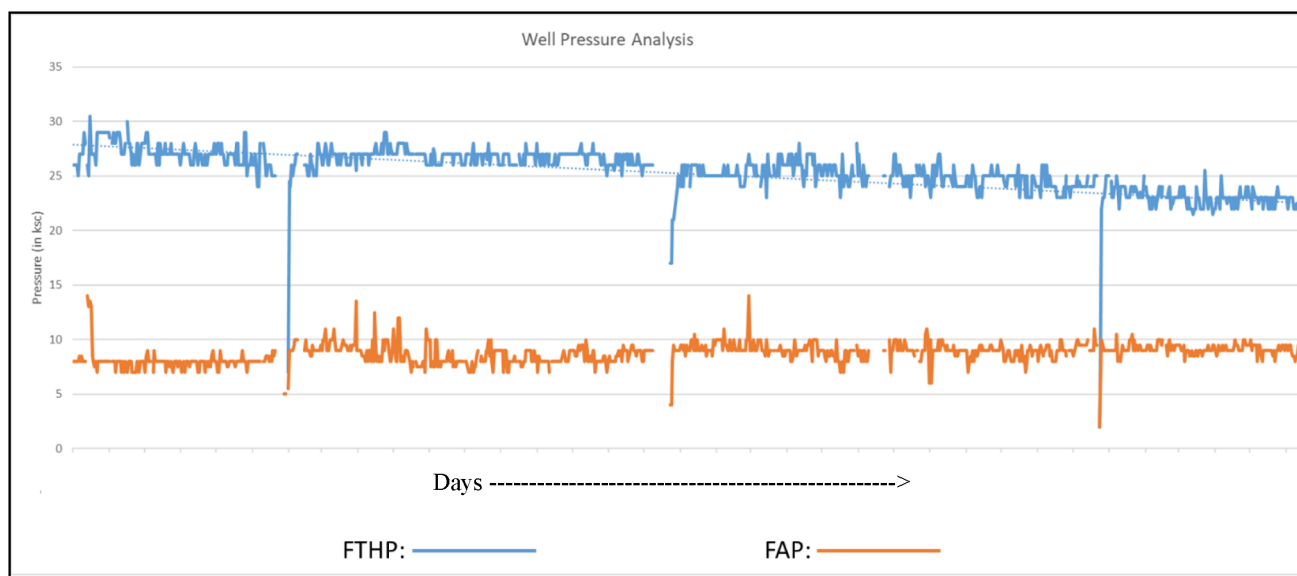


Figure 3—Representing the declining FTHP trend in Well M001

Manual drawdown enhancements were initially attempted through bean-up operations at defined intervals. While these interventions temporarily improved flow conditions, they posed several limitations:

- **High dependency on surface personnel**, resulting in inefficient resource utilization
- **Inconsistent results**, as liquid loading recurrence was unpredictable
- Operational difficulty in **timing interventions optimally**
- Risk of **sand production** if flow instability worsened

The cumulative effect of these challenges resulted in a **compromised vertical lift profile**, increased downtime, and gradual production decline. Given the open-hole completion, introducing conventional artificial lift systems or conducting coiled tubing unloading jobs was considered invasive and cost-prohibitive. The situation demanded a **simple, low-cost, autonomous solution** that could maintain optimal flow without recurring manual input or equipment overhauls.

Methodology and Intervention Design

Based on production behaviour observed in well M001, a detailed diagnostic study was undertaken to identify the root causes of flow instability. It was concluded that **accumulated emulsion blocks and segregated water** near the wellbore were increasing the density of the fluid column inside the tubing. This, combined with the well's low gas-oil ratio and back-pressure sensitivity, was leading to frequent liquid loading and **intermittent sluggish flow**.

Initial manual bean-up operations using periodic surface choke adjustments helped improve flow marginally as indicated by the flattening of FTHP trend in Fig-4, but they required **frequent human intervention**, offered inconsistent results, and were unsustainable for long-term optimization. These limitations prompted the design and implementation of a **low-cost, autonomous flow control mechanism**.

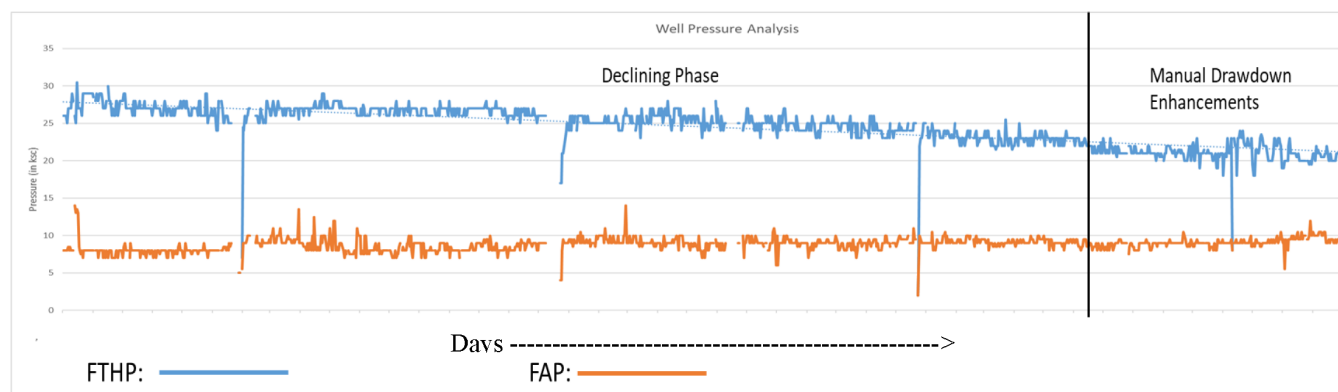


Figure 4—Impact on FTHP due to manual drawdown enhancement (Bean-up) operations in Well M001

A custom **Automated Tubing Purging Setup (ATPS)** was developed as per design shown in Fig-5 and installed at the wellhead. Fig-7 shows the first ATPS installed for M001. The system consisted of a **dual control valve assembly in two separate lines**, each having a pre-determined fixed choke (based on sensitivity analysis) installed ahead of control valve, converging into the common flowline, configured to operate based on predefined surface pressure thresholds:

- **Control Valve 1:** 8.75 mm Fixed Flow-Arm Bean (FAB), activated when surface pressure ≥ 22 kg/cm² [for normal condition] (Fig.-6 Case-1 shows the flow path indicated by bubbles in the active line segment during normal condition)
- **Control Valve 2:** 10 mm FAB, activated when surface pressure < 22 kg/cm² [for purging condition] (Fig.-6 Case-2 shows the flow path indicated by bubbles in the active line segment during purging operation)

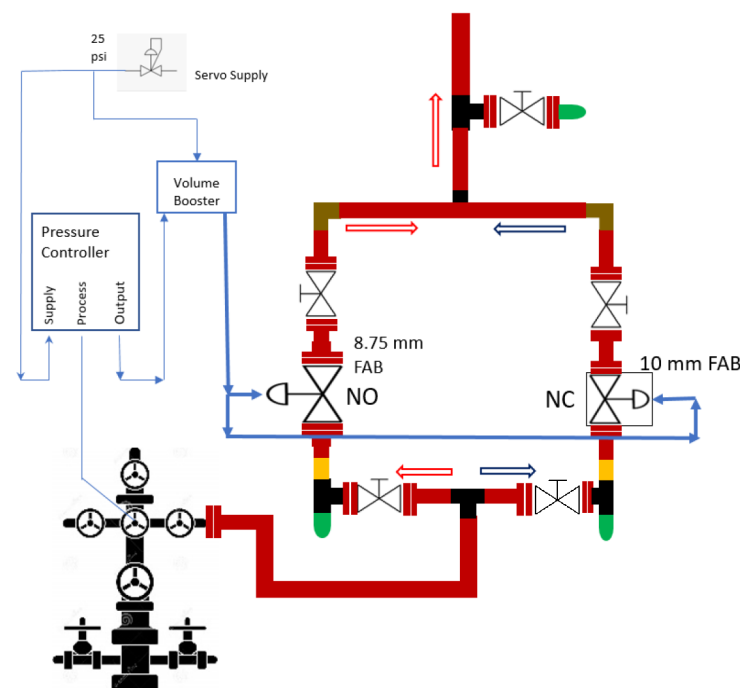


Figure 5—Construction Design of First Automated Tubing Purging Setup

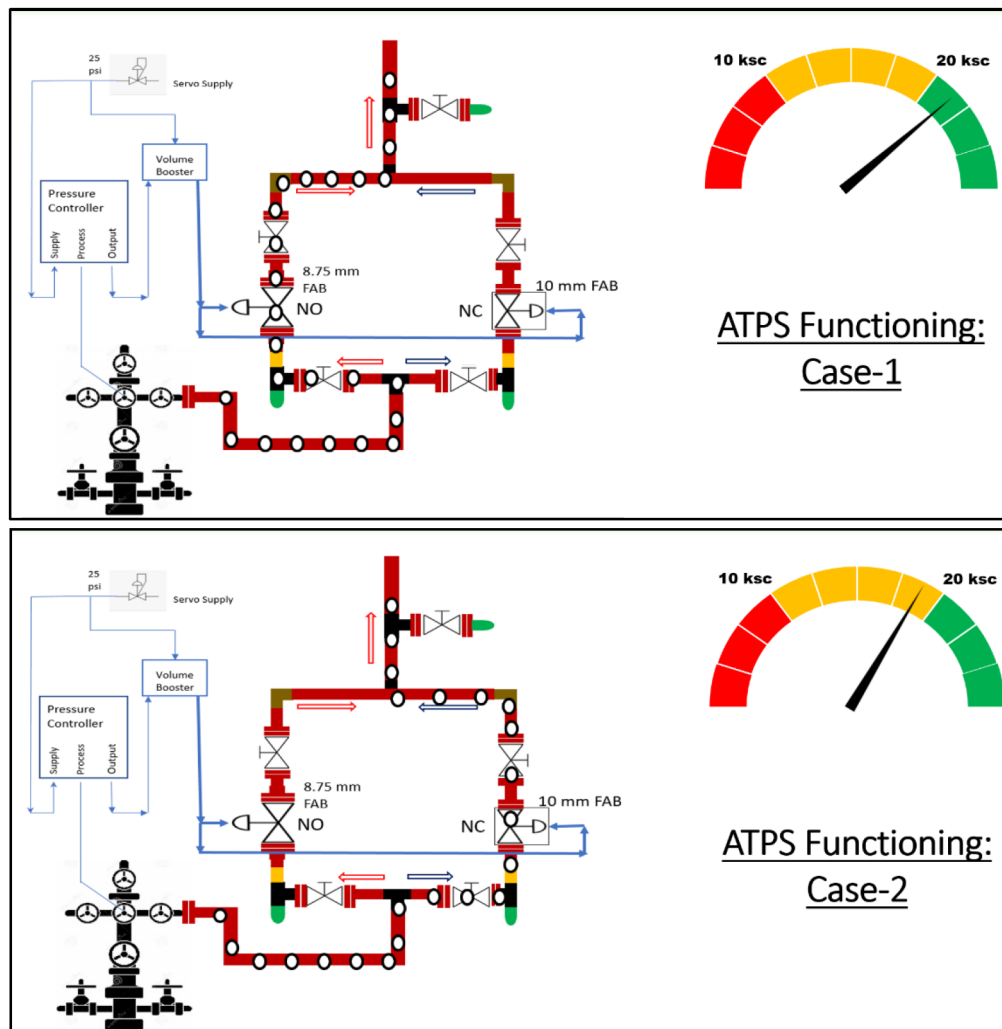


Figure 6—Automated Tubing Purging Setup Functioning Cases: (a) Case-1 Flow path during the Normal Flowing condition in the well. (b) Case-2 Flow path during the purging operation in the well.

This inverted operating mechanism allowed the well to **autonomously switch between normal and enhanced flow states** based on real-time conditions. The pressure drop resulting from bean-up triggered a shift in bottom-hole conditions (drop in tubing intake pressure), facilitating gas release from casing pressure and improving aeration within the tubing. This **lightened the multiphase column**, allowing accumulated emulsions and liquid blocks to be lifted from the near-wellbore region.

Once flow was stabilized and the tubing head pressure returned to the higher range, the system automatically reverted to the smaller bean, restoring the original flow configuration and ensuring production control without further manual input. This **dynamic switchover logic** formed the core of the ATPS, enabling a closed-loop, self-regulating production optimization system.



Figure 7—Automated Tubing Purging Setup installed in well M001

Results and Discussion

Following the installation of the Automated Tubing Purging Setup (ATPS) in well M001, significant and measurable improvements in production behaviour were observed. The intervention succeeded in stabilizing the flow regime, reducing the frequency of slugging, and restoring a favourable vertical lift profile.

Pressure Behaviour and Flow Stabilization

Flowing Tubing Head Pressure (FTHP), which had previously declined to 17-18 kg/cm² (occasionally touching 15-16 kg/cm²), showed a marked increase after the implementation of ATPS. Within months of installation, FTHP rose steadily to 63 -64 kg/cm² as evident from the graph in Fig-8, indicating improved lift dynamics. This increase is attributed to the automated bean-up cycle, which enabled periodic aeration from casing gas, reducing the density of the fluid column and lifting accumulated emulsions from the near-wellbore region.

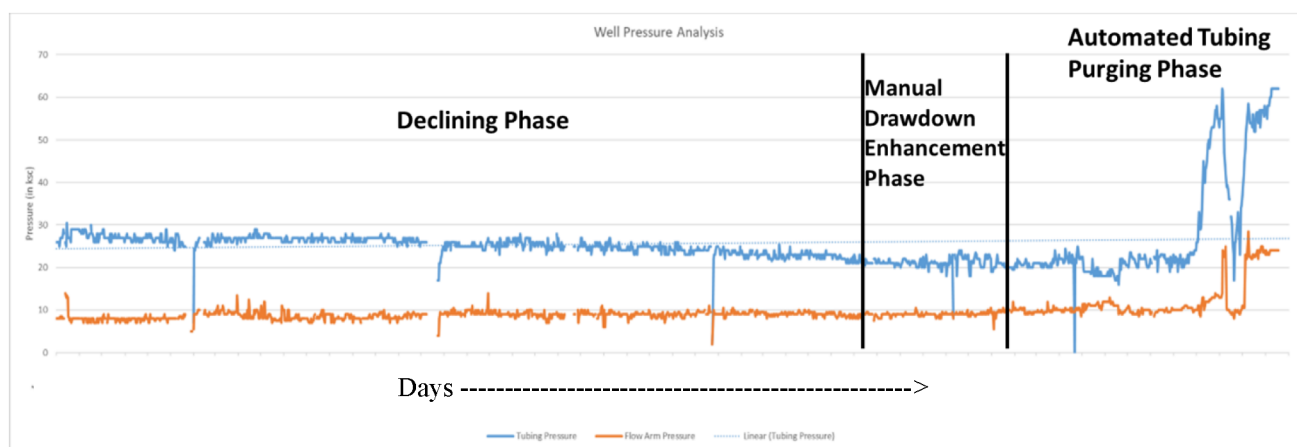


Figure 8—FTHP trends in M001 during the declining phase, manual drawdown enhancement phase and post-ATPS Phase.

As more emulsion blocks were unloaded from the wellbore and near well-bore region, the well gradually began to recover production characteristics similar to those observed nearly a decade earlier. Surface pressure analysis across multiple operational cycles showed improved pressure response and greater stability after each bean-up event. Downhole behaviour mirrored this trend. Fig-10 shows the various phases of a purging cycle with tubing intake pressure as the point of reference.

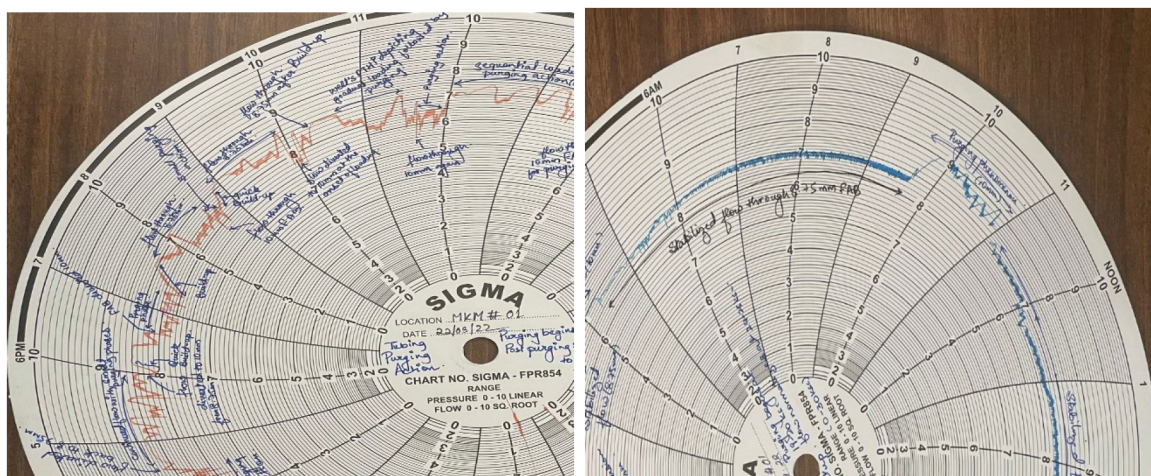


Figure 9—FTHP trends showing switching of flow-path in M001 post-ATPS installation. (a) Image-1 [Left]- FTHP behavior recorded on day-2 showing the increased number of flow-path changes. (b) Image-2 [Right]- FTHP behavior recorded on day-22 highlighting lesser dependency on purging setup.

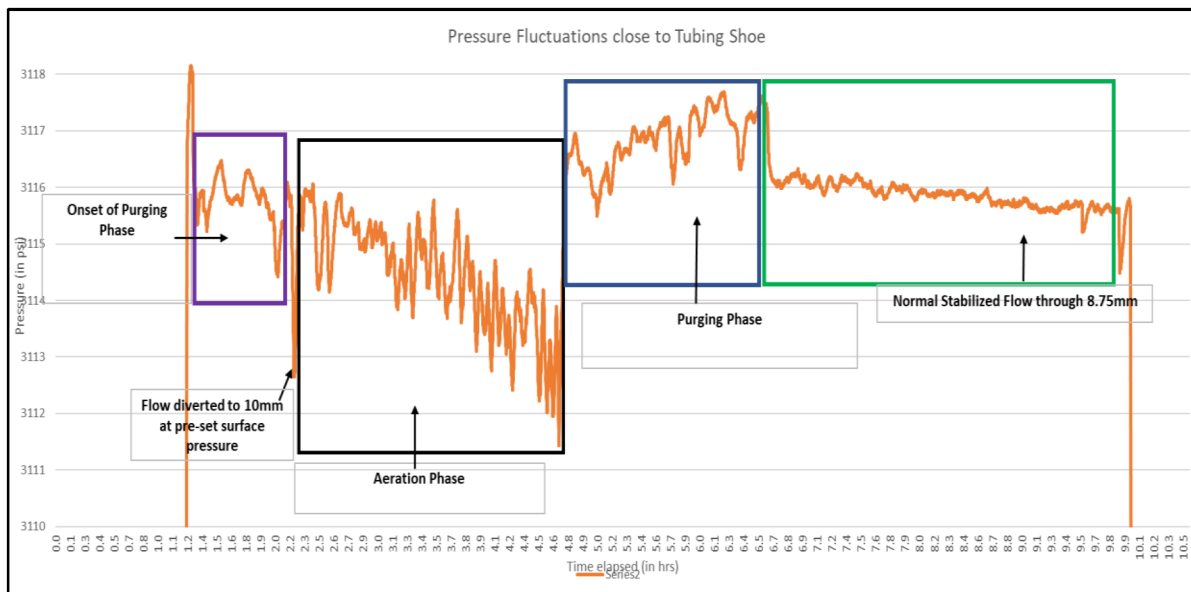


Figure 10—Tubing Intake Pressure behavior during a purging cycle

Notably, as the well continued to stabilize, its dependency on ATPS support diminished as shown in Fig-9 wherein Image-1 highlights frequent purging operation in day-2 and reduced dependency by day-22 shown in Fig-9. Eventually, it no longer required the purging system to sustain flow—an outcome indicative of long-term restoration of natural lift efficiency. The improved vertical lift performance led to an overall reduction of 40-42 kg/cm² pressure drop in the vertical regime owing to lighter multiphase fluid flow evident in Fig-11.

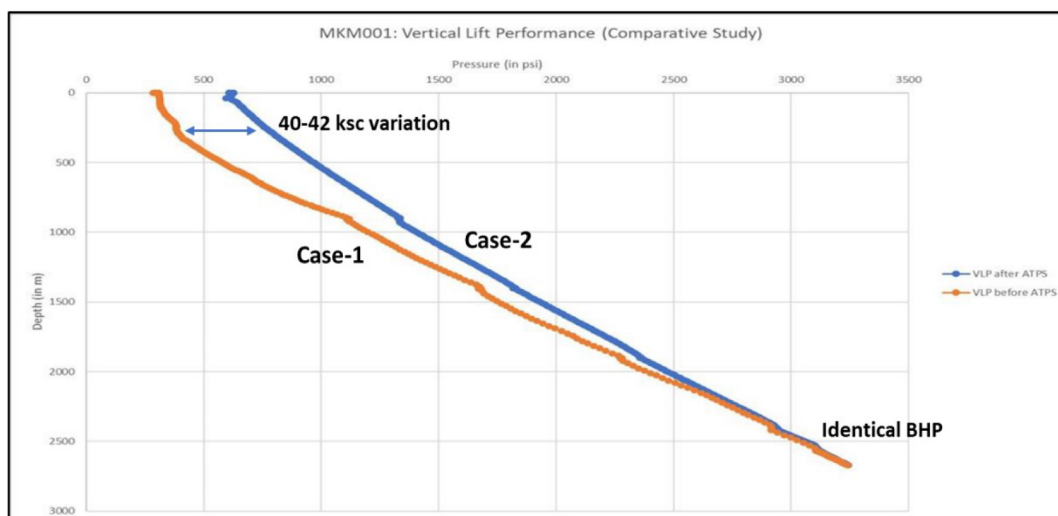


Figure 11—Vertical Lift Performance Comparison recorded before and after installation of ATPS

Production Performance

With improved aeration and stable lift conditions, total liquid production increased from 104 KLPD to 120 KLPD, maintaining 50% BS&W. This uplift corresponds to an incremental crude oil gain of approximately 8 KLPD. Simultaneously, daily natural gas production rose from 8,000 to 38,000 SCMD, reflecting a ~400% increase in gas output. These results validated the effectiveness of the automated cycling approach in enhancing both liquid and gas deliverability, without external energy input or artificial lift systems.

Economic Impact

The intervention was executed using in-house resources, with no additional capital expenditure for hardware or major modifications. Despite its simplicity, the system yielded significant economic returns:

- Incremental daily revenue from the well was approximately **5.5 times the total expenditure** incurred for setup and commissioning.
- **Monthly revenue gain** was estimated at **₹3.29 Crores**, factoring both the increase in crude oil and associated gas production.

A consolidated summary of field performance and economic indicators is presented in [Table 1](#).

Table 1—Summary of Key Performance Indicators Before and After ATPS Implementation (Well M001)

Parameter	Before ATPS	After ATPS	Change / Benefit
Flowing Tubing Head Pressure (FTHP)	17–18 kg/cm ²	63–64 kg/cm ²	↑ ~3.5x pressure improvement
Liquid Production Rate	104 KLPD	120 KLPD	↑ ~15% overall increase
Crude Oil Production Rate	~52 KLPD	~60 KLPD	↑ ~8 KLPD incremental gain
Water Cut (BS&W)	~50%	~50%	Stable
Natural Gas Production Rate	8,000 SCMD	38,000 SCMD	↑ ~400% uplift
Manual Bean-Up Frequency	Frequent	Eliminated	✓ Operational automation
Flow Stability / Slugging	Intermittent, unstable	Stable and continuous	✓ Flow regime stabilized
Downhole Pressure Response	Poor	Improved communication	✓ Improved inflow behavior
Energy / Lift Input Required	None	None	✓ No external energy added
Surface Setup Requirements	Conventional bean setup	Dual bean with control	✓ Minor surface modification only
Instrumentation Used	Manual gauge reading	Basic pressure switch	✓ No digital telemetry needed
Additional CAPEX	Not feasible	₹1.97 Lakhs (in-house)	✓ Fully cost-optimized
Daily Revenue Gain (Est.)	–	₹10.9 Lakhs/day	↑ ~5.5× intervention cost in daily return
Monthly Revenue Gain (Est.)	–	₹3.29 Crores	✓ Based on oil and gas uplift
Post-Implementation Dependency	Continuous intervention	Independent, self-sustained	✓ System phased out over time

Operational Observations

Beyond production and pressure improvements, the setup proved highly effective in resolving heading and fluid loading while significantly reducing the need for manual intervention. By automating what was previously a labor-intensive drawdown enhancement process, both operational efficiency and field safety were improved. The system demonstrated high reliability in realtime valve switching based on surface pressure thresholds and remained responsive to dynamic flow conditions throughout the operational period.

Scalability and Field Deployment

Encouraged by the success of the Automated Tubing Purging Setup (ATPS) in well M001, the system was subsequently replicated in **three additional wells: M002, M025, and M027** [Summarized results of the implementation in [Table-2](#)]. These wells exhibited similar production challenges, including heading, liquid loading, and sensitivity to back-pressure, though each presented unique operational constraints in terms of flow profile, and surface layout.

Table 2—Summarized results highlighting pre- and post-ATPS installation in candidate well

Well	Pre-ATPS Condition	Post-ATPS Condition	Key Benefit
M002	Frequent manual bean-ups, unstable flow	Stabilized FTHP @ 22–24 kg/cm ²	Eliminated manual intervention
M025	Moderate slugging	Steadier flow, uplifted liquid rate	Sustained self-flow
M027	Severe water loading, low GOR	Stabilized pressure with modified setup	Enabled autonomous unloading

In each case, the ATPS configuration was **customized to suit local well conditions** while retaining its core design philosophy: a **dual-bean control setup** governed by **surface pressure-triggered logic**. The primary objective remained consistent — to enable **periodic, autonomous drawdown** that unloads emulsion blocks and improves vertical lift performance without requiring artificial lift systems or continuous surface monitoring.

M002 and M027

- In **M002**, the well had previously required frequent manual bean-ups to maintain minimal flow. After ATPS deployment, the declining tubing head pressure trend was arrested and stabilized continuous self-flow @ 22-24 kg/cm², thus, eliminating manual intervention.
- **M027**, a horizontal well with semi-closed completion, was initially constrained by low productivity due to **severe water loading** combined with **low GOR**. Following ATPS installation, the pressure profile stabilized. For this well, a modified configuration as shown in Fig-12 was implemented: one flowline contained a fixed choke for normal production, while the secondary control valve, triggered at low FTHP, enabled enhanced flow via a **dual flow path** to facilitate liquid unloading.

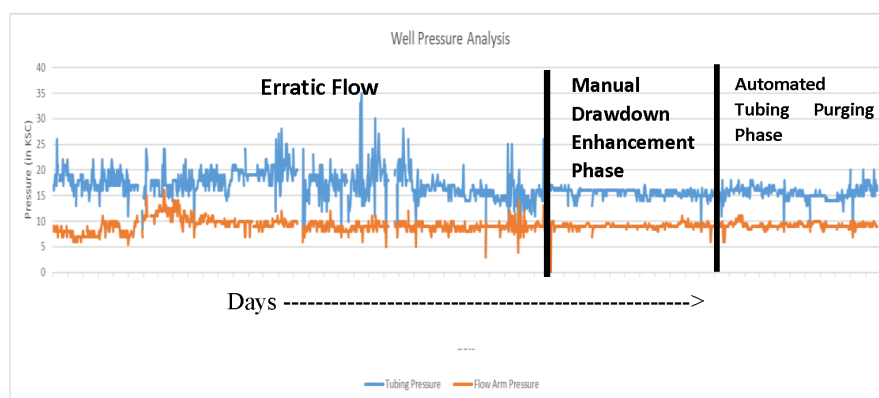


Figure 12—(a) ATPS installed in M027 (b) FTHP behavior recorded in M027 during pre/post-ATPS installation

Both wells demonstrated **notable improvements in flow continuity**, reduced downtime, and a substantial reduction in operational burden. The system preserved its **fail-safe and passive architecture**, requiring only basic pressure instrumentation and retrofitted surface bean loop modifications — all accomplished using **minimal inventory**.

Replication Criteria

The following criteria were considered for evaluating the suitability of ATPS deployment:

- Evidence of **intermittent flow** caused by liquid loading or heading
- Sufficient **annular gas pressure** to support tubing aeration (*applicable for open-hole completions*)
- **Surface accessibility** for bean-loop modifications
- **Economic viability** based on anticipated production uplift relative to setup cost

The consistent results across multiple wells confirmed the solution's **technical scalability**, **cost efficiency**, and **operational simplicity**. Its non-invasive nature, compact footprint, and reliance on existing surface infrastructure make it a **highly applicable intervention** for mature brownfield assets where reservoir energy is declining and artificial lift solutions are either technically infeasible or economically unjustifiable.

Lessons Learned

The deployment of the Automated Tubing Purging Setup (ATPS) across multiple mature wells provided several valuable insights into production behaviour, intervention design, and long-term well management:

Simplicity Can Outperform Complexity

Despite its basic surface-level design and reliance on existing infrastructure, the ATPS yielded outcomes comparable to more advanced artificial lift systems in terms of flow stabilization, pressure recovery, and production uplift. This underscores the principle that **solutions need not be technologically complex to be effective**—especially in brownfield scenarios where **cost sensitivity and adaptability** are critical.

Timely Diagnostics Are Key

The success of the intervention hinged on a **detailed diagnosis of flow behaviour** and emulsion dynamics. Recognizing the cyclic nature of liquid loading, low GOR, and heading phenomena enabled a targeted solution that directly addressed the root cause, rather than relying on conventional methods with limited success in such wells.

Controlled Bean-Up Works Better When Automated

Manual drawdown operations, while effective to some extent, introduced inconsistency and dependency on field personnel. The **automation of bean-up cycles using pressure triggers** not only enhanced operational safety and reliability but also allowed the well to "self-regulate," eventually restoring stable flow without continued intervention.

Scalability Is a Function of Logic, Not Layout

Each of the replicated wells had unique characteristics — ranging from horizontal geometry to semi-closed completions. Yet, the **core logic of pressure-governed dual-path switching remained applicable**, proving that **scalability is rooted more in logic design than physical well layout**, provided diagnostic conditions are favourable.

Restoring Flow Can Be a Transitional Process

In M001, it was observed that the ATPS was **initially essential to sustain production**, but over time, as emulsions were unloaded and aeration improved, **the system became redundant**. This highlights that well optimization can sometimes be **transitional rather than continuous**, with interventions helping wells re-attain a more stable natural lift regime.

Conclusions

The successful implementation of a simple yet effective **Automated Tubing Purging Setup (ATPS)** in a set of mature wells has demonstrated that **low-cost, field-adapted interventions** can significantly enhance production performance and operational continuity in brownfield environments.

The ATPS serves as a practical model for **lean optimization** in late-life wells—enhancing vertical lift efficiency, minimizing operational risk, and extending the productive life of existing assets. As the industry continues to seek value from maturing reservoirs, **simple, replicable, and intelligent field solutions like this** offer a compelling path toward sustained production and operational excellence.